

A Negative Answer to a Question on Monotone Maximal Chains in Finite Groups

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Abstract

Monakhov and Sokhor asked whether every finite group has an unrefinable subgroup chain whose successive indices are nondecreasing. We give a negative answer. For every odd prime p , we construct a soluble matrix group $G_p \leq \mathrm{GL}_5(p)$ of order $2^6 p^8$ with no such chain. In particular, G_3 has order 419904. The construction is a small instance of an example of Kohler; the obstruction is certified by three elementary maximal-subgroup calculations.

1 Introduction

Let $G \neq 1$ be finite. An unrefinable subgroup chain

$$1 = G_0 < G_1 < \cdots < G_n = G$$

is one in which G_{i-1} is maximal in G_i for every i . Associate to the chain the indices $j_i = [G_i : G_{i-1}]$. Following Monakhov and Sokhor [2], call the chain increasing if

$$j_1 \leq j_2 \leq \cdots \leq j_n.$$

They asked whether every finite group contains an increasing unrefinable chain. We prove the following.

Theorem 1. *For every odd prime p , there is a soluble group G_p of order $2^6 p^8$ which has no increasing unrefinable subgroup chain.*

The smallest group in the family has order $2^6 3^8 = 419904$. Our construction is the $t = 2$ case of the construction in Kohler's work on maximal-subgroup indices [1]. Kohler used it to show that a group whose maximal subgroups have prime or prime-square index can contain subgroups with arbitrarily large maximal-subgroup index. Here we use the nested index pattern itself.

2 A lemma on maximal subgroups

For a finite group A , write

$$\mathcal{I}(A) = \{[A : B] : B \text{ is maximal in } A\}.$$

Lemma 2. *Let V be an elementary abelian p -group, let H be a p' -group acting on V , and put $K = V \rtimes H$. Every maximal subgroup of K is of one of the following forms:*

1. $V \rtimes J$, up to the natural identification, where J is maximal in H ;
2. a V -conjugate of $W \rtimes H$, where W is a maximal proper H -submodule of V .

Proof. Let T be maximal in K . If $V \leq T$, the claim follows on passing to $K/V \cong H$. Suppose $V \not\leq T$. Then $VT = K$. Put $W = V \cap T$. The group T normalizes W , while V centralizes it, so $W \triangleleft K$. In K/W , the groups T/W and WH/W are complements to V/W . Schur–Zassenhaus makes these complements conjugate by an element of V/W . Thus T is V -conjugate to $W \rtimes H$, and maximality of T is equivalent to maximality of W among proper H -submodules of V . $\square$

In particular, if H is a 2-group, subgroups of the first type have index 2. If V is completely reducible, subgroups of the second type have indices equal to the orders of the irreducible quotients of V .

3 The counterexample family

Fix an odd prime p and put $F = \mathbb{F}_p$. Let

$$s = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad t = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad D = \langle s, t \rangle \leq \mathrm{GL}_2(F).$$

Then $D \cong D_8$, where D_8 has order 8. The natural FD -module $U = F^2$ is absolutely irreducible. Indeed, over an algebraic closure the only t -invariant lines are the two coordinate lines, and s interchanges them.

Let $H = D \times D$. Define

$$L = \left\{ \ell(x, y, z) = \begin{pmatrix} I_2 & x & z \\ 0 & 1 & y \\ 0 & 0 & I_2 \end{pmatrix} : x \in F^{2 \times 1}, y \in F^{1 \times 2}, z \in F^{2 \times 2} \right\}.$$

Here the block sizes are 2, 1, 2. Direct multiplication gives

$$\ell(x, y, z)\ell(x', y', z') = \ell(x + x', y + y', z + z' + xy'). \quad (1)$$

Thus $|L| = p^8$. Put

$$Z = \{\ell(0, 0, z)\}, \quad X = \{\ell(x, 0, 0)\}, \quad Y = \{\ell(0, y, 0)\}.$$

The group Z is central in L . The outer products xy span $F^{2 \times 2}$, so (1) gives $L' = Z$. Also L has exponent p : explicitly,

$$\ell(x, y, z)^p = \ell\left(0, 0, pz + \binom{p}{2}xy\right) = 1.$$

Consequently

$$\Phi(L) = L^p L' = Z. \quad (2)$$

Embed H in $\mathrm{GL}_5(F)$ through the block-diagonal matrices $\mathrm{diag}(A, 1, B)$, with $A, B \in D$. Conjugation acts by

$$x \mapsto Ax, \quad y \mapsto yB^{-1}, \quad z \mapsto AzB^{-1}.$$

It therefore normalizes L . Set

$$G_p = L \rtimes H \leq \mathrm{GL}_5(p).$$

We have $|G_p| = p^8 \cdot 8^2 = 2^6 p^8$. Both L and H are soluble, so G_p is soluble.

As an H -module,

$$L/Z = \overline{X} \oplus \overline{Y}, \quad (3)$$

where $\overline{X}$ and $\overline{Y}$ are nonisomorphic irreducible modules of dimension 2. The first factor of $D \times D$ acts naturally on $\overline{X}$ and the second acts naturally (dually) on $\overline{Y}$. Moreover,

$$Z \cong U \boxtimes U^* \quad (4)$$

is the external tensor product for the two direct factors of H . Hence Z is an irreducible H -module of dimension 4 (after scalar extension, irreducible modules for a direct product are external tensor products).

4 The index obstruction

Lemma 3. *The following maximal-index spectra occur:*

$$\begin{aligned} \mathcal{I}(G_p) &= \{2, p^2\}, \\ \mathcal{I}(M_X) &= \mathcal{I}(M_Y) = \{2, p^2, p^4\}, \\ \mathcal{I}(N) &= \{2, p^4\}, \end{aligned}$$

where

$$M_X = (Z \times X) \rtimes H, \quad M_Y = (Z \times Y) \rtimes H, \quad N = Z \rtimes H.$$

Every maximal subgroup of G_p of index p^2 is conjugate to M_X or M_Y . Every maximal subgroup of M_X or M_Y of index p^2 is conjugate to N .

Proof. First, every maximal subgroup M of G_p contains Z . If not, then $G_p = ZM$, and the modular law and (2) give

$$L = L \cap ZM = Z(L \cap M) = \Phi(L)(L \cap M).$$

The defining property of the Frattini subgroup forces $L \cap M = L$, which contradicts $Z \not\leq M$.

It follows that the maximal subgroups of G_p are the inverse images of the maximal subgroups of

$$G_p/Z = (\overline{X} \oplus \overline{Y}) \rtimes H.$$

Apply Lemma 2 and (3). Maximal subgroups coming from H have index 2, and those coming from either irreducible summand have index p^2 . This proves the first spectrum and the assertion about M_X, M_Y .

In M_X , the normal elementary abelian subgroup $Z \times X$ is a direct sum of irreducible H -modules of dimensions 4 and 2. Another application of Lemma 2 gives indices $2, p^2, p^4$; the index- p^2 case removes the X summand and gives a conjugate of N . The same argument applies to M_Y . Finally, Z is irreducible of dimension 4, so Lemma 2 applied to $N = Z \rtimes H$ gives $\mathcal{I}(N) = \{2, p^4\}$. $\square$

Proof of Theorem 1. Suppose

$$1 = G_0 < G_1 < \cdots < G_n = G_p$$

is unrefinable and its indices $j_i = [G_i : G_{i-1}]$ are nondecreasing. By Lemma 3, $j_n \in \{2, p^2\}$. We cannot have $j_n = 2$: then every j_i would be 2, whereas p divides $|G_p|$. Thus $j_n = p^2$, and G_{n-1} is conjugate to M_X or M_Y .

Now $j_{n-1} \leq p^2$. The middle spectrum in Lemma 3 shows that j_{n-1} is 2 or p^2 . It cannot be 2, for the same order argument applied to G_{n-1} . Hence $j_{n-1} = p^2$, and G_{n-2} is conjugate to N .

Finally, $j_{n-2} \leq p^2$, but $\mathcal{I}(N) = \{2, p^4\}$. Therefore $j_{n-2} = 2$. This forces all preceding indices to equal 2, impossible because p divides $|N|$. The contradiction proves the theorem. $\square$

Remark 4. For $p = 3$, a GAP verification computes the successive spectra as

$$\{2, 9\}, \quad \{2, 9, 81\}, \quad \{2, 81\},$$

and independently exhausts all admissible branches. The code is available in the repository accompanying this note.

References

- [1] J. Kohler, *Finite groups with all maximal subgroups of prime or prime square index*, Canad. J. Math. **16** (1964), 435–442. doi:10.4153/CJM-1964-046-6.
- [2] V. S. Monakhov and I. L. Sokhor, *On indices of maximal chains in finite groups*, Results Math. **80** (2025), article 155. doi:10.1007/s00025-025-02468-5.